Syntax and kinding

Named notation for the de Bruijn formalization. Fresh binders and capture-avoiding substitution are understood.

$$\begin{aligned} K, L &::= * \mid K \Rightarrow L \\ A, B, F &::= X \mid \mathrm{Top} \mid A \to B \mid \forall X <: A :: K.\, B \mid \lambda X.\, A \mid F\, A \\ t, u &::= x \mid \lambda x : A.\, t \mid t\, u \mid \Lambda X <: A :: K.\, t \mid t[A]_K \\ \Gamma &::= \varnothing \mid \Gamma, X <: A :: K \\ \Delta &::= \varnothing \mid \Delta, x : A \end{aligned}$$

Operator-lambda kinds are carried by judgments. Term-level type abstraction and application retain their kind annotations.

$$\mathrm{Top}_{*} := \mathrm{Top} \qquad \mathrm{Top}_{K \Rightarrow L} := \lambda X.\, \mathrm{Top}_{L} \qquad \Gamma, X :: K \;:=\; \Gamma, X <: \mathrm{Top}_{K} :: K$$

Type-variable contexts

$$\frac{}{\vdash \varnothing\ \mathsf{wf}}\ \textsc{C-Empty} \qquad \frac{\vdash \Gamma\ \mathsf{wf} \qquad \Gamma \vdash A :: K}{\vdash \Gamma, X <: A :: K\ \mathsf{wf}}\ \textsc{C-Bind}$$

Kinding

Kinding reads only the kinds in Γ, independently of bounds.

$$\frac{(X <: U :: K) \in \Gamma}{\Gamma \vdash X :: K}\ \textsc{K-Var} \qquad \frac{}{\Gamma \vdash \mathrm{Top} :: *}\ \textsc{K-Top} \qquad \frac{\Gamma \vdash A :: * \qquad \Gamma \vdash B :: *}{\Gamma \vdash A \to B :: *}\ \textsc{K-Arrow}$$

$$\frac{\Gamma, X :: K \vdash A :: L}{\Gamma \vdash \lambda X.\, A :: K \Rightarrow L}\ \textsc{K-Abs} \qquad \frac{\Gamma \vdash A :: K \qquad \Gamma, X <: A :: K \vdash B :: *}{\Gamma \vdash \forall X <: A :: K.\, B :: *}\ \textsc{K-All}$$

$$\frac{\Gamma \vdash F :: K \Rightarrow L \qquad \Gamma \vdash A :: K}{\Gamma \vdash F\, A :: L}\ \textsc{K-App}$$

Type conversion

$$(\lambda X.\, A)\, B \longrightarrow_{\beta} A[B/X]$$

Reduction is closed under all type constructors. Its reflexive, symmetric, transitive closure is $\equiv_{\beta}$. There is no η rule.

Formal definitions: `theories/Syntax.v; theories/HigherOrder/Subtyping.v`

Declarative subtyping

The judgment $\Gamma \vdash A <: B :: K$ compares types at the same kind. These rules are used with well-formed contexts.

$$\frac{\Gamma \vdash A :: K}{\Gamma \vdash A <: A :: K}\ \text{S-Refl} \qquad \frac{\Gamma \vdash A :: K}{\Gamma \vdash A <: \mathrm{Top}_K :: K}\ \text{S-Top}$$

$$\frac{(X <: A :: K) \in \Gamma \qquad \Gamma \vdash A :: K}{\Gamma \vdash X <: A :: K}\ \text{S-Var} \qquad \frac{\Gamma \vdash A <: B :: K \qquad \Gamma \vdash B <: C :: K}{\Gamma \vdash A <: C :: K}\ \text{S-Trans}$$

$$\frac{\Gamma \vdash A' <: A :: * \qquad \Gamma \vdash B <: B' :: *}{\Gamma \vdash A \to B <: A' \to B' :: *}\ \text{S-Arrow}$$

$$\frac{\Gamma \vdash A' <: A :: K \qquad \Gamma, X <: A' :: K \vdash B <: B' :: *}{\Gamma \vdash (\forall X <: A :: K.\, B) <: (\forall X <: A' :: K.\, B') :: *}\ \text{S-All}$$

$$\frac{\Gamma, X <: \mathrm{Top}_K :: K \vdash A <: B :: L}{\Gamma \vdash \lambda X.\, A <: \lambda X.\, B :: K \Rightarrow L}\ \text{S-Abs} \qquad \frac{\Gamma \vdash F <: F' :: K \Rightarrow L \qquad \Gamma \vdash C :: K}{\Gamma \vdash F\, C <: F'\, C :: L}\ \text{S-App}$$

$$\frac{\Gamma \vdash A :: K \qquad \Gamma \vdash B :: K \qquad A \equiv_\beta B}{\Gamma \vdash A <: B :: K}\ \text{S-Conv}$$

Full quantification and pointwise operators

S-All permits different bounds. The bound is contravariant, and the bodies are compared under the right-hand bound A'. S-Abs uses an unbounded operator parameter. S-App uses the same argument C on both sides; no general covariance rule for operator arguments is present.

S-Trans permits arbitrary well-kinded intermediate types. The erasure used in the conservativity proof is not an object-language conversion rule.

Explicitly annotated typing

Γ contains type-variable bounds; Δ contains term-variable types. The result type in $\Gamma; \Delta \vdash t : A$ has kind $*$.

$$\frac{(x : A) \in \Delta \qquad \Gamma \vdash A :: *}{\Gamma; \Delta \vdash x : A}\ \text{T-Var} \qquad \frac{\Gamma \vdash A :: * \qquad \Gamma; \Delta, x : A \vdash t : B}{\Gamma; \Delta \vdash \lambda x : A.\, t : A \to B}\ \text{T-Abs}$$

$$\frac{\Gamma; \Delta \vdash t : A \to B \qquad \Gamma; \Delta \vdash u : A}{\Gamma; \Delta \vdash t\, u : B}\ \text{T-App}$$

$$\frac{\Gamma \vdash A :: K \qquad \Gamma, X <: A :: K; \Delta \vdash t : B}{\Gamma; \Delta \vdash \Lambda X <: A :: K.\, t : \forall X <: A :: K.\, B}\ \text{T-TAbs}$$

$$\frac{\Gamma; \Delta \vdash t : \forall X <: A :: K.\, B \qquad \Gamma \vdash C <: A :: K}{\Gamma; \Delta \vdash t[C]_K : B[C/X]}\ \text{T-TApp}$$

$$\frac{\Gamma; \Delta \vdash t : A \qquad \Gamma \vdash A <: B :: *}{\Gamma; \Delta \vdash t : B}\ \text{T-Sub}$$

In T-TAbs, X is fresh for the existing contexts. The named notation keeps Δ unchanged; the de Bruijn implementation shifts its stored types.

The old fragment

$$S, T ::= X \mid \text{Top} \mid S \to T \mid \forall X <: S :: *.\, T$$

Old contexts bind only base-kind type variables with old bounds. Old terms use base-kind type abstraction and application, with old lambda annotations and old type arguments. The proved conservativity results compare judgments whose contexts, terms, and endpoints satisfy these restrictions.

The formalization specifies kinding, subtyping, and typing. It does not define an operational semantics for terms.

Formal definitions: `theories/Terms.v; theories/HigherOrder/Typing.v`